

Disproof of Conjecture 00000000057: Prime Cycle Lengths in Minimum-Degree-Three Graphs

TLMC submission

September 29, 2026

Abstract

We disprove Conjecture 00000000057, which asserts that every n -vertex graph of minimum degree at least 3 contains $\Omega(\log n / \log \log n)$ cycles of pairwise distinct prime lengths. The counterexample is the disjoint union of any number of copies of $K_{3,3}$: it has minimum degree 3 on $n = 6k$ vertices, yet all of its cycle lengths lie in $\{4, 6\}$, so the number of distinct prime cycle lengths is 0 for every k , while $\log n / \log \log n \rightarrow \infty$.

1 The conjecture

Conjecture 1 (00000000057). *Every n -vertex graph of minimum degree at least 3 contains $\Omega(\log n / \log \log n)$ cycles of pairwise distinct prime lengths.*

2 The counterexample family

For $k \geq 1$ let G_k denote the disjoint union of k copies of the complete bipartite graph $K_{3,3}$. Thus G_k has $n = 6k$ vertices and $9k$ edges.

Proposition 1. $\delta(G_k) = 3$ for every $k \geq 1$.

Proof. In $K_{3,3}$ every vertex is adjacent to all 3 vertices of the opposite part and to no vertex of its own part, so every vertex has degree exactly 3. Taking a disjoint union does not alter any vertex degree. Hence $\delta(G_k) = 3 \geq 3$, and $|V(G_k)| = 6k$ is unbounded as $k \rightarrow \infty$. $\square$

Lemma 2. *Every cycle of G_k has length 4 or 6.*

Proof. Every cycle of a disjoint union of graphs lies entirely inside one connected component, i.e. inside one copy of $K_{3,3}$: consecutive vertices of a cycle are adjacent, and there are no edges between distinct copies. So it suffices to consider $K_{3,3}$ with bipartition $A \dot{\cup} B$, $|A| = |B| = 3$. A cycle in a bipartite graph alternates between the two parts, so every cycle has even length $2m$ and uses exactly m vertices of A and m vertices of B . Since $|A| = |B| = 3$ we have $m \leq 3$, and since $K_{3,3}$ is simple (no parallel edges) a cycle has $m \geq 2$. Hence $2m \in \{4, 6\}$, and both lengths are realized. $\square$

Lemma 3. *No cycle of G_k has prime length.*

Proof. By Lemma 2 every cycle length is $4 = 2 \cdot 2$ or $6 = 2 \cdot 3$, both composite. $\square$

3 The disproof

Theorem 4. *Conjecture 00000000057 is false.*

Proof. Let G_k be the family above and write $n = |V(G_k)| = 6k$. By Proposition 1, G_k has minimum degree at least 3 for every k , and $n \rightarrow \infty$. By Lemmas 2 and 3, G_k contains no cycle of prime length at all, so the number of cycles of pairwise distinct prime lengths is 0 for every k .

Since $\log n / \log \log n \rightarrow \infty$ as $n \rightarrow \infty$, for every constant $c > 0$ there are arbitrarily large k with $c \cdot \log(6k) / \log \log(6k) > 0$ (indeed the right-hand side is positive for all $n \geq 3$ and unbounded). The asserted lower bound $0 \geq c \cdot \log n / \log \log n$ therefore fails for every $c > 0$: the number of distinct prime cycle lengths is not $\Omega(\log n / \log \log n)$. $\square$

Remark 5. *Already for $k = 1$, i.e. $n = 6$, the graph $K_{3,3}$ has $\lfloor \log_2 6 \rfloor / \lfloor \log_2 (\lfloor \log_2 6 \rfloor + 1) \rfloor = 2/1 = 2 > 0$ as a discrete stand-in for the target growth rate, while its prime-length cycle count is 0: exactly nine 4-cycles and six 6-cycles (up to reversal), no others.*

Remark 6. *The obstruction is that a minimum-degree condition alone cannot control component structure: the conjecture would need a connectivity hypothesis (or a bound on the number of components) to rule out this family. For connected graphs of minimum degree 3 the conjecture is not addressed here.*

4 Verification artifacts

- `reproduce.py` brute-force enumerates *all* simple cycles of G_k for $k \in \{1, 2, 3, 5, 8\}$ directly from the adjacency definition and confirms: minimum degree 3, all cycle lengths in $\{4, 6\}$, zero distinct prime lengths.
- `lean4/` contains a full Lean 4 (core, no Mathlib, v4.33.1) formalization with zero axioms and zero `sorry`: every cycle of the family has even length at least 3 (hence is non-prime), every vertex has three pairwise distinct neighbors, and the vertex count $6k$ is unbounded.